

High-z galaxy candidates in the COSMOS-Web field– F150W dropout

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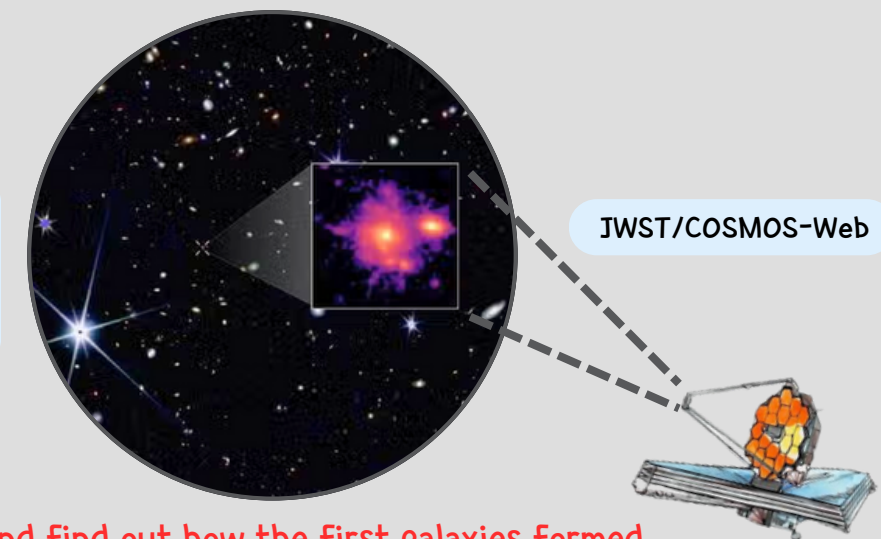
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Revealing Early Galaxies and Protoclusters with JWST

- Understanding when and how the first galaxies formed is key to understanding the early Universe. Characterizing the physical properties of early galaxies, such as the **UV Luminosity Function (UVLF)** and **Star Formation Rate Density (SFRD)**, reveals their evolutionary pathways
- The large-area coverage (0.54 deg²) of the JWST/COSMOS-Web survey allows us to systematically probe extremely distant ($z > 12$) galaxy populations with improved statistical confidence.

Population of high red-shift galaxies
 $z > 12$

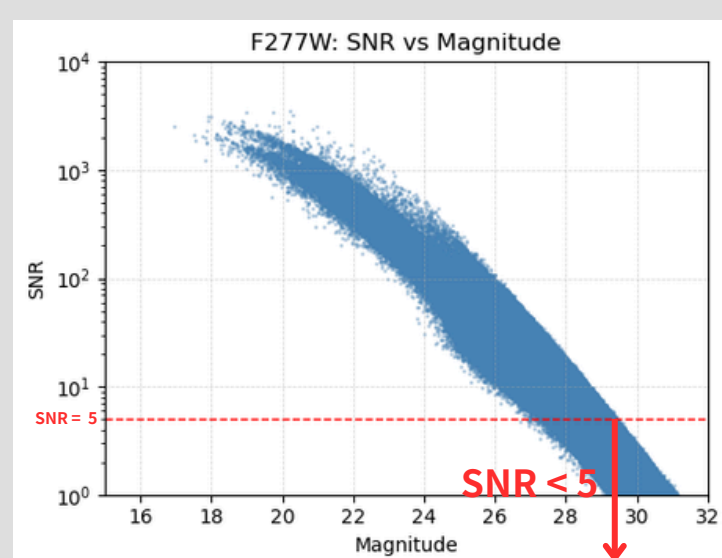


Look into the early universe and find out how the first galaxies formed.

How to select objects?

1. SNR cut

Remove objects with SNR < 5, an object will be kept as long as its SNR exceeds 5 in at least one filter. (F277W for example)

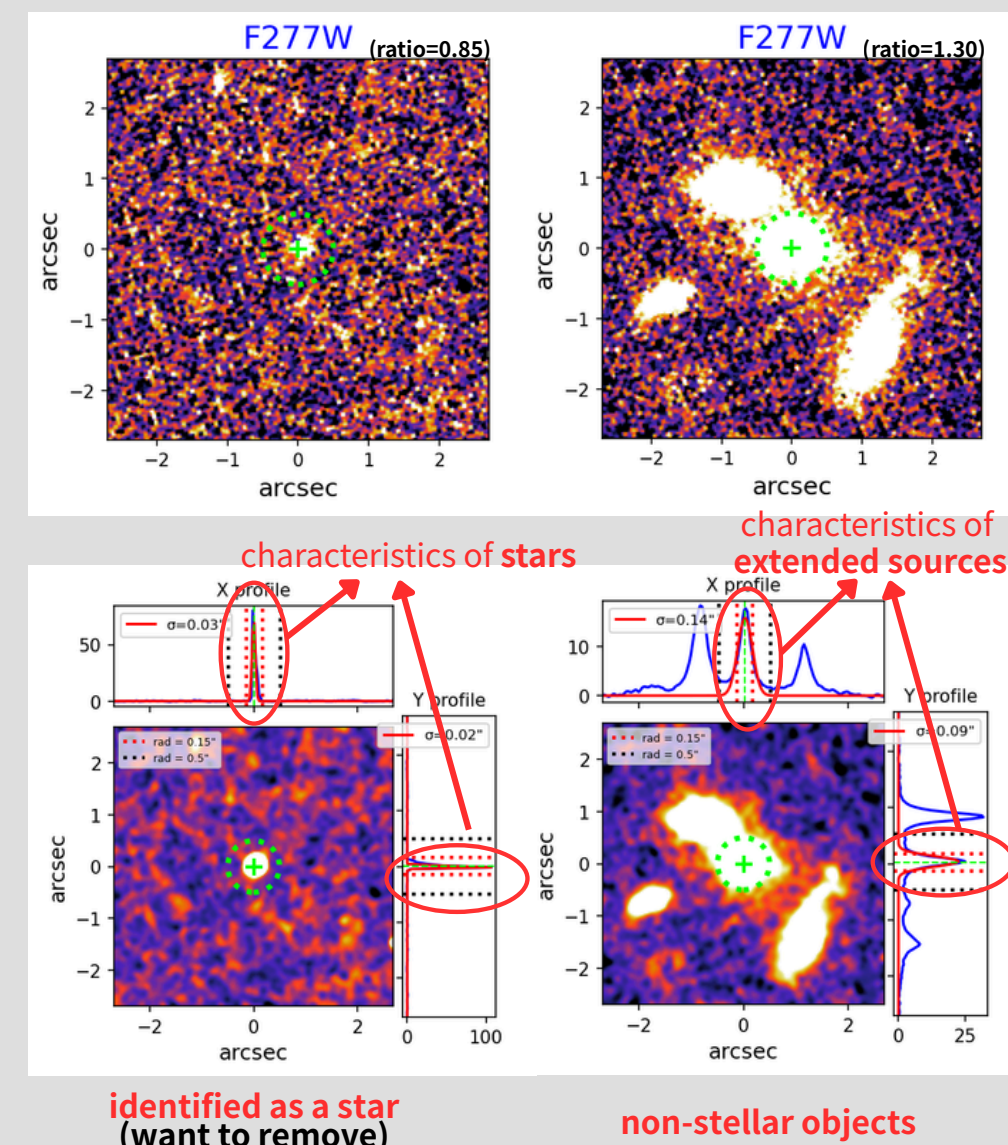
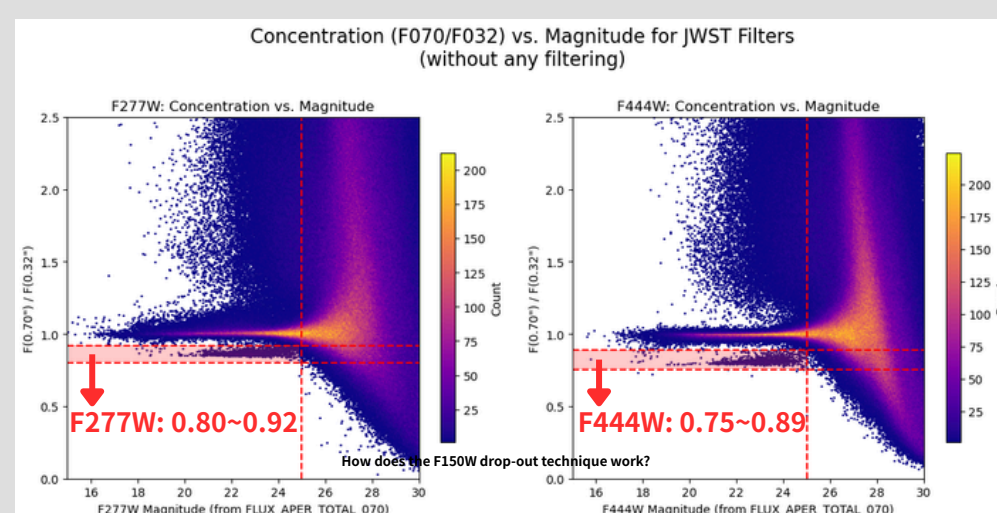


2. COSMOS crossmatch

Remove objects matching the COSMOS2020 catalog.

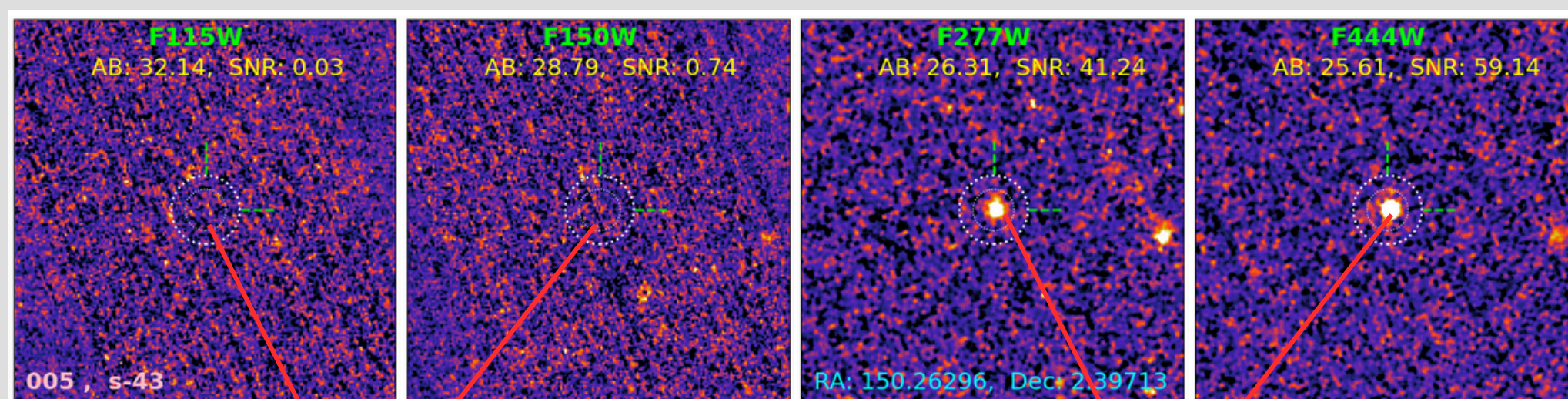
3. Ratio selection

The ratio as a function of magnitude was plotted for the F277W and F444W filters, and the point-size ratio—derived from point and extended sources measured through different apertures—was used to define the region characteristic of stellar objects.



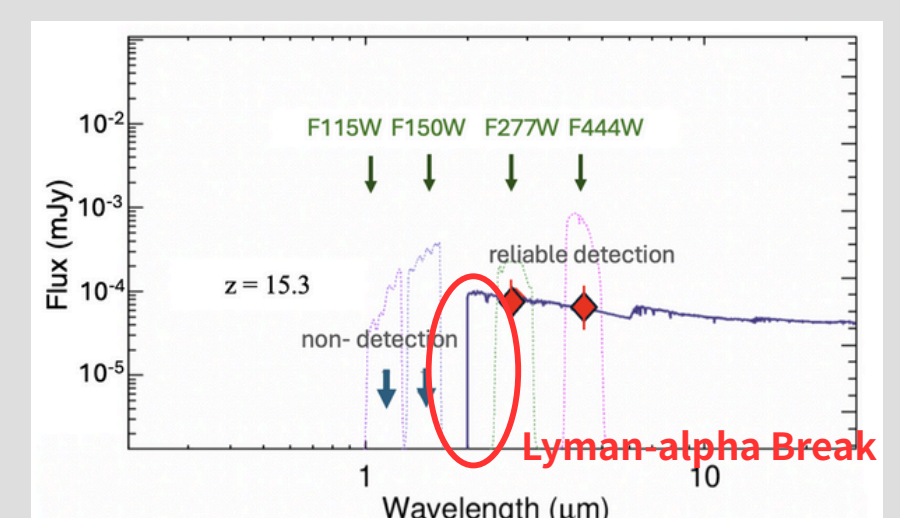
How does the F150W drop-out technique work, and why is the 150 filter used?

- High-redshift galaxies show a sharp drop in brightness shortward of 1216 Å due to **absorption of ultraviolet light by neutral hydrogen**.
- For galaxies at very high redshifts ($z \gtrsim 12$), this “**break**” is **redshifted into the near-infrared**, where JWST/NIRCam can observe it.
- If a galaxy disappears (“drops out”) in F150W but appears in redder filters (F277W, F444W) → it’s a strong high-redshift candidate.



disappears in F115W, F150W bands

appears in F277W, F444W bands



Next Steps

- Select and confirm high-redshift galaxy candidates from the filtered catalog. Based on literature estimates, approximately **375 candidates** are expected to be identified across the whole sky.
- Compute the **UV luminosity function** and **star formation rate density** for the selected high-redshift galaxy candidates.

Insights

1. Early Galaxy Population Statistics

- The number and properties of high-redshift galaxy candidates provide **constraints on the abundance and distribution of early galaxies in the universe**.
- This helps refine models of galaxy formation during the first billion years.

2. UV Luminosity Function (UV LF)

- The UV LF reveals the **intrinsic brightness distribution** of early galaxies.
- Comparing with theoretical predictions can test **galaxy formation models and cosmic reionization scenarios**.

3. Star Formation Rate Density (SFRD)

- The derived SFRD indicates **how rapidly stars were forming in the early universe**.
- It provides insight into the **timing and efficiency of early star formation**, informing the epoch of reionization studies.